SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B. Tech AI&ML (2024-2028)	**Date:**	14/08/2026

Code of Conduct

I T. Bhanu Pavan Varma (192425380) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

Q1. Semantic Representation in Customer Support Chatbots

The chatbot represents customer requests using action–object structures such as ACTIVATE(Roaming, Customer) and QUERY(DataBalance, Customer).

1. Analyze the semantic representations and identify action-object relationships.

The semantic representations correctly identify the **action** and the **object/service** involved in each request.

Customer Query	Action	Object
Activate international roaming	ACTIVATE	Roaming
Deactivate caller tune	DEACTIVATE	CallerTune
Check data balance	QUERY	DataBalance
Enable 5G service	ACTIVATE	5GService

For example:

ACTIVATE(Roaming,Customer)
means that the customer wants the **Roaming service to be activated**.

QUERY(DataBalance,Customer)
represents a request to **retrieve the customer's data-balance information**.

Thus, the semantic representation converts natural-language queries into a structured form that the chatbot can process.

2. Determine which query contains semantic interpretation errors.

From the given table:

Query	Actual Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	**Activate Caller Tune**
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Q2 contains the semantic interpretation error.

The actual intent is:

DEACTIVATE(CallerTune,Customer)
but the chatbot predicts:

ACTIVATE(CallerTune,Customer)
The object **CallerTune** is correctly identified, but the action is incorrectly interpreted as **ACTIVATE instead of DEACTIVATE**.

Therefore, the semantic error is specifically an **action classification error**.

3. Evaluate how meaning representation affects chatbot decision accuracy.

Meaning representation has a direct effect on chatbot accuracy because it converts the user's sentence into an actionable semantic structure.

For example:

"Deactivate caller tune service"
↓
DEACTIVATE(CallerTune, Customer)
↓
Deactivate caller-tune service

If the representation is correct, the chatbot can execute the appropriate operation. If the action is incorrectly represented, the system may perform the opposite operation.

In this case, Q2 could cause a serious service error because the customer wants to **deactivate** a service but the chatbot may **activate** it instead.

Thus, accurate semantic representation is essential for reliable decision-making.

4. Recommend improvements for telecom customer support.

The system can be improved by:

1. Using **intent classification** to distinguish actions such as activate, deactivate, and query.
2. Using contextual information to distinguish similar requests.
3. Maintaining a domain-specific telecom vocabulary.
4. Using entity recognition for services such as roaming, caller tune, and 5G.
5. Validating the predicted action before executing an important service operation.
6. Using feedback from incorrect predictions such as Q2 to retrain the semantic model.

Conclusion

The semantic representation system performs correctly for **Q1, Q3, and Q4**, while **Q2 demonstrates an action-level semantic error**. Improving action recognition and contextual understanding would increase chatbot decision accuracy.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $Active(x) \rightarrow Producing(x)$
- $Produces(x,y) \wedge Active(x) \rightarrow Available(y)$
- $Maintenance(x) \rightarrow \neg Producing(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

Q2. First-Order Predicate Calculus for Smart Manufacturing

The manufacturing plant contains four machines:

- M1 – Active
- M2 – Active
- M3 – Maintenance
- M4 – Active

The rules are:

$Active(x) \rightarrow Producing(x)$ $Produces(x,y) \wedge Active(x) \rightarrow Available(y)$ $Maintenance(x) \rightarrow \neg Producing(x)$
These facts and rules are given in the case study.

1. Represent the production data using First-Order Predicate Calculus.

The machine states can be represented as:

$Active(M1)$ $Active(M2)$ $Maintenance(M3)$ $Active(M4)$
Using the first rule:

$$Active(x) \rightarrow Producing(x)$$

we derive:

$$Producing(M1)\ Producing(M2)\ Producing(M4)$$

From:

$$Maintenance(M3)$$

and:

$$Maintenance(x) \rightarrow \neg Producing(x)$$

we derive:

$$\neg Producing(M3)$$

Therefore, M1, M2, and M4 are producing, while M3 is not producing.

2. Analyze which products are currently available.

The relevant rule is:

$$Produces(x,y) \wedge Active(x) \rightarrow Available(y)$$

This means a product y is available when a machine x produces it and that machine is active.

However, the given case **does not provide any specific Produces(machine, product) facts**.

For example, there is no information such as:

$$Produces(M1,Gear)$$

Therefore, the exact products currently available **cannot be determined from the supplied data**.

What we can conclude is:

Any product produced by M1, M2, or M4 would be available because these machines are active.

3. Infer whether Gear production is affected by maintenance.

The case tells us:

$$\text{Maintenance}(M3)$$

Therefore:

$$\neg\text{Producing}(M3)$$

So M3 cannot currently produce anything.

However, the data does **not state that M3 produces Gear**.

4. Evaluate predicate logic in industrial decision-support systems.

Predicate logic is useful because it converts machine conditions into formal rules and allows automatic inference.

For example:

$\text{Active}(M1) \rightarrow \text{Producing}(M1)$
allows the system to infer that M1 is producing without explicitly storing that fact.

Similarly:

$\text{Maintenance}(M3) \rightarrow \neg\text{Producing}(M3)$
immediately identifies M3 as unavailable for production.

This makes predicate logic useful for:

- Machine monitoring
- Production planning
- Fault detection
- Resource availability
- Automated decision support

However, the system depends on complete and accurate facts. The missing Produces(machine, product) relationship in this case demonstrates that **logical inference cannot generate information that is not present in the knowledge base**.

Conclusion

Predicate calculus provides a precise framework for industrial reasoning, but its conclusions are limited by the completeness of the available production data.

3. Word Sense Disambiguation in E-Commerce Search Engines.

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

Q3. Word Sense Disambiguation in E-Commerce Search Engines

The search system contains ambiguous words such as **Apple, Mouse, Java, and Python**. The search-click information provides contextual evidence for identifying their intended meanings.

1. Determine the correct word sense for each query.

Query	Correct Sense	Evidence
Apple accessories	**Technology/Brand**	iPhone Charger
Mouse wireless	**Computer Device**	Bluetooth Mouse
Java tutorial	**Programming Language**	Coding Lessons
Python course	**Programming Language**	Software Development Training

Apple accessories

Although **Apple** can refer to a fruit or technology brand, the clicked result is **iPhone Charger**.

Therefore:

$$\text{Apple} = \text{Technology/Brand}$$

Mouse wireless

The word "mouse" can refer to an animal or computer device. The clicked result is **Bluetooth Mouse**.

Therefore:

$$\text{Mouse} = \text{Computer Device}$$

Java tutorial

Java can refer to an island or programming language. The result is **Coding Lessons**.

Therefore:

$$\text{Java} = \text{Programming Language}$$

Python course

Python can refer to a snake or programming language. The result is **Software Development Training**.

Therefore:

$$\text{Python} = \text{Programming Language}$$

2. Identify semantic cues used for disambiguation.

The major cues are:

Context words

Words such as:

accessories
wireless
tutorial
course

provide information about the intended meaning.

Clicked results

The user's selected result provides strong evidence:

iPhone Charger → Apple technology
Bluetooth Mouse → computer mouse
Coding Lessons → Java programming
Software Development Training → Python programming

Domain context

Because the system is an **e-commerce search engine**, product-related meanings are more likely than unrelated meanings.

Thus, WSD can combine query context, clicked results, and domain information.

3. Impact of incorrect sense selection

Incorrect word-sense selection can significantly reduce search quality.

For example, if:

Java tutorial

is interpreted as an island-related query, the search engine may return travel information instead of programming tutorials.

Similarly, interpreting:

Mouse wireless

as an animal query would produce completely irrelevant results.

This can cause:

- Lower search relevance
- Poor recommendations
- Reduced click-through rate
- Poor customer experience
- Incorrect product ranking

Therefore, accurate WSD directly improves search performance.

4. Industrial-scale WSD strategy

A practical WSD system should combine multiple sources of evidence:

1. **Contextual embeddings** to understand surrounding words.
2. **Query classification** to identify the user's domain.
3. **Search history and clicked results** as behavioral evidence.
4. **Knowledge bases** containing word senses and entities.
5. **User personalization** when appropriate.
6. **Continuous learning** from user clicks and feedback.

For example:

Python course
↓
Context = course
↓
Domain = programming
↓
Python = Programming Language
↓
Recommend coding courses

Conclusion

The given search logs show that contextual information and clicked results successfully resolve the ambiguity of all four words. A large-scale WSD system should combine linguistic context with behavioral and domain information to improve recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1.Analyze how syntactic structures contribute to semantic role assignment.

2.Determine whether the assigned semantic roles are appropriate.

3.Evaluate potential errors that could arise from incorrect parsing.

4.Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Q4. Syntax-Driven Semantic Analysis in Healthcare

The healthcare system processes sentences using **Subject–Verb–Object** structures and assigns semantic roles such as Agent, Instrument, Recipient, and Symptom.

1. Analyze how syntactic structures contribute to semantic role assignment.

Syntax provides information about the grammatical relationship between words.

For example:

Doctor prescribed medicine to patient.

The syntactic structure is:

Subject	Verb	Object
Doctor	prescribed	medicine

This helps identify:

- Doctor → Agent
- Medicine → object/instrument
- Patient → Recipient

Similarly:

Patient reported severe headache.

The structure helps identify:

- Patient → Agent/Experiencer
- Headache → Symptom

Thus, syntactic parsing provides the structural information required for semantic role assignment.

2. Determine whether the assigned semantic roles are appropriate.

The given semantic roles are:

Entity	Assigned Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient

Headach e	Symptom

Doctor → Agent

This is appropriate because the doctor performs the action of prescribing.

Patient → Recipient

This is appropriate in:

Doctor prescribed medicine to patient.

The patient receives the medicine.

Headache → Symptom

This is appropriate because the headache represents the patient's medical symptom.

Medicine → Instrument

This is the questionable assignment.

In:

Medicine reduced blood pressure.

Medicine is better interpreted as the **Cause/Agent-like entity** responsible for the reduction, rather than an Instrument used by an intentional agent.

Therefore, the role assignment is **mostly appropriate**, but **Medicine → Instrument** could be refined depending on the semantic-role scheme being used.

3. Potential errors from incorrect parsing

Incorrect syntactic parsing can produce serious semantic errors in healthcare NLP.

For example, confusing the subject and object could change:

Doctor prescribed medicine to patient.

into an incorrect interpretation where the patient appears to prescribe the medicine.

This could lead to errors in:

- Patient records
- Medication extraction
- Treatment history
- Clinical decision support
- Medical question answering

4. Methods to improve syntax-driven semantic analysis

Healthcare NLP systems can be improved using:

1. Domain-specific parsing

Train models on medical documents so they understand terminology such as:

hypertension
medication
headache
blood pressure

2. Dependency parsing

Dependency parsing can identify precise relationships between subjects, verbs, objects, and modifiers.

3. Semantic Role Labeling

SRL can identify roles such as:

Agent
Patient
Recipient
Cause
Symptom

4. Context-aware models

Transformer-based language models can use surrounding context to resolve ambiguity.

5. Medical knowledge bases

Clinical ontologies and medical knowledge bases can validate extracted relationships.

6. Human validation

Important clinical information can be checked by medical professionals or validated through trusted clinical rules.

Conclusion

Syntax-driven semantic analysis provides the structural foundation for understanding medical sentences. However, syntax alone may not always distinguish roles such as **Instrument and Cause**, so combining dependency parsing, semantic-role labeling, medical knowledge, and contextual models can improve accuracy and reliability.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________